

Python for Data Science - Syllabus

Module 1: Python Basics

- What is Python?
- Installing Python & Jupyter Notebook
- Writing your first Python program
- Variables and Data Types
- Basic Input and Output
- Operators (e.g., +, -, *, /)

Module 2: Programming Logic

- Conditional Statements (if, else)
- Loops (for, while)
- Functions (basic)
- Simple Error Handling

Module 3: Python Data Structures

- Lists
- Tuples
- Dictionaries
- Sets
- Using loops with these structures

Module 4: Functions

- Types of functions.
- Types of Arguments.
- Writing programs Using Functions

Module 5: OOPS Concepts

- Basics
- Inheritance, Encapsulations.....

Module 6: Advanced Concepts

- File Handling
- Regular Expressions

- Decorators
- Generators and Iterators

Module 7:Frame Works

- Flask, Django Basics

Module 8: Introduction to NumPy

- What is NumPy?
- Arrays vs Lists
- Basic Array Operations
- Array Indexing & Slicing

Module 9: Introduction to Pandas

- What is Pandas?
- Reading CSV files
- Understanding DataFrames
- Selecting Columns and Rows
- Simple Filtering & Sorting